

Supplementary Material for “Reliability Failure of Remaining-Useful-Life Prediction Intervals under Sensor Degradation”

Yi Wang

Every table below is generated by script from the named source file in the released repository (`jeff-1980/rul-interval-reliability`, tag `v1.2-submission`); no value is transcribed by hand. All perturbations are seeded from SHA-256 digests; with `PYTHONHASHSEED=0` and deterministic cuDNN, two independent runs on the same machine reproduced 71 of the 74 result files bit-for-bit and the remaining three in every field except a wall-clock timing field; the files are listed with their hashes in `REPRODUCE.md`.

S0. PROVENANCE OF EVERY TABLE

Each main-text and appendix table, with the generating script in `code/` of the released repository, the label version (“clipped” is the 125-cycle target clip), the evaluation unit, whether engines are equally weighted, how seeds and perturbation realisations are aggregated, and the data over which perturbation scales are referenced.

Table	Script / result file		Labels	Unit	Engine-weighted	Aggregation	Noise range
Protocol 2x2 (Table I)	build_table1.py (re-inference over all four cells' existing checkpoints under this release's protocol); underlying checkpoints from train_lstm.py (V/F) and protocol_2x2_quadram4.py (T/F); the T/W and V/W cells' checkpoints come from the superseded pre-audit training scripts under code/superseded/, and are shipped as fixed weights in this release		min(official_RUL-1,125)	terminal	yes	mean of 5 seeds; ECE on pooled seeds	–
Clean benchmark (Table II)	eval_nll_clean_b1.py, eval_splitcp_b1.py, eval_mcdropout_b1.py, eval_fd003_mechanism_b1.py (re-evaluate existing checkpoints under this release's test-truth convention, no re-training); fair_calibration_main_table.py (fixed variance, MC Dropout); cost_table_lstm.py; cost_table_transformer.py; table2_clean_merge.py (others); eval_*_per_engine_coverage.py (per-engine column); eval_*_deep_ensemble.py (ensemble rows)		min(official_RUL-1,125)	terminal (per-engine: trajectory)	yes	mean of 5 seeds; ECE per seed	–
Same-partition ensemble (Table III)	eval_samesplit_ensemble_b1.py		min(official_RUL-1,125)	terminal	yes	one partition (seed 42), five initialisations	–
Attribution, LSTM (Table IV)	attribution_interp_lstm.py; clamp fraction from clamp_frac.py		min(official_RUL-1,125)	terminal	yes	seed-pooled predictions; interpolated crossover	sensor channels only on FD002/FD004 (15 of 18 feature columns; the 3 operating-condition settings are left unperturbed); official training file for FD001/FD003
Structured degradation (Table V)	run_sweep_degradation_transformer.py with sweep_engine.py (both backbones), half_life_degradation.py		min(official_RUL-1,125)	terminal	yes	5 models x 5 realisations, grand mean	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Drift controls (Table VI)	drift_controls.py; drift_bias_equal_magnitude_control.py; drift_fixed_endpoint_shuffle_control.py		min(official_RUL-1,125)	terminal	yes	5 models; deterministic except shuffled/bias sign draws	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Decision stress test (Table VII)	maintenance_decision_one_sided.py; unclipped RUL from maintenance_stress_test_rul_correction_v2.py		min(official_RUL-1,125) for triggers; unclipped RUL reported as official_RUL-1	terminal	yes	5 models x 5 realisations	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Pre-flight diagnostics (not in main tables)	diagnostic_perturbation_target_a1.py; alignment_a2.py (see results/diagnostics/)	diagnostic_target_	both official_RUL and min(official_RUL-1,125) compared side by side (A2); sensors-only/settings-only/joint compared side by side (A1)	terminal	yes	5 seeds	A1 compares sensor-only vs settings-only vs joint on FD002/FD004; A2 does not perturb features
Fair calibration (Appendix B)	fair_calibration_sigma_fixed.py		min(official_RUL-1,125)	terminal	yes	mean of 5 seeds	–

Table	Script / result file	Labels	Unit	Engine-weighted	Aggregation	Noise range
Attribution, Transformer (Appendix C)	attribution_interp_transformer.py	min(official_RUL-1,125)	terminal	yes	seed-pooled predictions; interpolated crossover	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Seeds and ensemble size (Appendix D)	eval_lstm_ensemble_n5_vs_n15.py; independent_replication.py	ensemble_size_and_ min(official_RUL-1,125) for ensemble_size_and_independent_replication.py only; predecessor convention for eval_lstm_ensemble_n5_vs_n15.py	terminal	yes	15 seeds; three disjoint ensembles	–
Latency (Appendix E)	latency_lstm.py; latency_transformer.py	–	–	–	median and IQR over 11 repetitions	–

S1. FOUR-WAY ATTRIBUTION AT THE INTERPOLATED TEN-POINT-LOSS POINT

Source: `results/attribution/attribution_lstm_exact_interp.json` and `results/attribution/attribution_transformer_exact_interp.json`. C_{ij} is coverage with mean output from condition i and scale output from condition j (0 clean, 1 perturbed). Order A varies the mean first; order B varies the scale first. Interaction = $(C_{11} - C_{10}) - (C_{01} - C_{00})$, computed from the four coverages; it is positive in all sixteen cells and reaches +0.043.

TABLE II
FOUR-WAY ATTRIBUTION, BOTH BACKBONES, AT THE INTERPOLATED POINT (f_{oob} IN %).

Backbone	Data	Mech.	f_{oob}	C_{00}	C_{10}	C_{01}	C_{11}	μ (A)	σ (A)	μ (B)	σ (B)	Int.	Both
LSTM	FD001	Heterosc.	0.55	0.864	0.792	0.826	0.764	-0.0717	-0.0283	-0.0625	-0.0375	+0.0092	yes
LSTM	FD001	CP-norm	0.85	0.878	0.806	0.842	0.778	-0.0716	-0.0284	-0.0644	-0.0356	+0.0072	yes
LSTM	FD002	Heterosc.	0.87	0.913	0.822	0.883	0.813	-0.0911	-0.0089	-0.0696	-0.0304	+0.0215	yes
LSTM	FD002	CP-norm	0.82	0.932	0.846	0.903	0.832	-0.0857	-0.0143	-0.0712	-0.0288	+0.0145	yes
LSTM	FD003	Heterosc.	0.24	0.858	0.762	0.831	0.758	-0.0964	-0.0036	-0.0730	-0.0270	+0.0234	yes
LSTM	FD003	CP-norm	0.63	0.896	0.806	0.868	0.796	-0.0902	-0.0098	-0.0719	-0.0281	+0.0183	yes
LSTM	FD004	Heterosc.	1.17	0.885	0.779	0.848	0.785	-0.1058	+0.0058	-0.0631	-0.0369	+0.0427	yes
LSTM	FD004	CP-norm	1.19	0.877	0.771	0.843	0.777	-0.1064	+0.0064	-0.0664	-0.0336	+0.0400	yes
Transformer	FD001	Heterosc.	0.89	0.880	0.800	0.860	0.780	-0.0800	-0.0200	-0.0797	-0.0203	+0.0002	yes
Transformer	FD001	CP-norm	0.63	0.892	0.810	0.867	0.792	-0.0823	-0.0177	-0.0751	-0.0249	+0.0071	yes
Transformer	FD002	Heterosc.	0.79	0.892	0.803	0.861	0.792	-0.0886	-0.0114	-0.0690	-0.0310	+0.0195	yes
Transformer	FD002	CP-norm	0.79	0.911	0.823	0.879	0.811	-0.0884	-0.0116	-0.0675	-0.0325	+0.0210	yes
Transformer	FD003	Heterosc.	0.11	0.870	0.770	0.849	0.770	-0.0998	-0.0002	-0.0790	-0.0210	+0.0208	yes
Transformer	FD003	CP-norm	0.26	0.876	0.783	0.859	0.776	-0.0935	-0.0065	-0.0832	-0.0168	+0.0102	yes
Transformer	FD004	Heterosc.	0.45	0.842	0.745	0.820	0.742	-0.0968	-0.0032	-0.0781	-0.0219	+0.0187	yes
Transformer	FD004	CP-norm	0.44	0.850	0.755	0.827	0.750	-0.0957	-0.0043	-0.0763	-0.0237	+0.0194	yes

Cross-seed standard deviations of C_{10} and C_{11} at the grid point nearest the crossover (`results/attribution/attribution_raw_per_seed_nearest_grid.json`) lie between 0.011 and 0.081; the percentage contributions in the main text are computed from seed-pooled predictions (aggregate-then-difference), not as a mean of per-seed ratios.

Paired bootstrap of mean dominance. For each cell, $|\Delta_\mu| - |\Delta_\sigma|$ was computed per seed at the same interpolated ten-point-loss point as the table above (per-seed C_{ij} interpolated between the same bracketing levels), separately for each decomposition order, and bootstrapped over the five seeds (2000 resamples). The intervals reflect variability across the five training-and-partition seeds only; they are conditional on the fixed test set, the seeded perturbation realisations and the chosen interpolated crossover, and do not propagate those sources of uncertainty. Source: `results/attribution/attribution_bootstrap_by_order_exact_interp.json`.

Backbone	Data	Mech.	Mean first: mean [95% CI]	> 0	Scale first: mean [95% CI]	> 0
LSTM	FD001	Heterosc.	+0.043 [+0.029, +0.058]	yes	+0.025 [+0.009, +0.039]	yes
LSTM	FD001	CP-norm	+0.043 [+0.028, +0.061]	yes	+0.029 [+0.006, +0.051]	yes
LSTM	FD002	Heterosc.	+0.082 [+0.070, +0.094]	yes	+0.039 [+0.028, +0.051]	yes
LSTM	FD002	CP-norm	+0.071 [+0.046, +0.096]	yes	+0.042 [+0.028, +0.057]	yes
LSTM	FD003	Heterosc.	+0.087 [+0.063, +0.113]	yes	+0.043 [+0.032, +0.055]	yes
LSTM	FD003	CP-norm	+0.078 [+0.039, +0.116]	yes	+0.044 [+0.027, +0.061]	yes
LSTM	FD004	Heterosc.	+0.096 [+0.086, +0.107]	yes	+0.029 [+0.017, +0.046]	yes
LSTM	FD004	CP-norm	+0.100 [+0.084, +0.120]	yes	+0.028 [+0.020, +0.037]	yes
Transformer	FD001	Heterosc.	+0.060 [+0.036, +0.084]	yes	+0.058 [+0.052, +0.062]	yes
Transformer	FD001	CP-norm	+0.065 [+0.036, +0.095]	yes	+0.048 [+0.030, +0.066]	yes
Transformer	FD002	Heterosc.	+0.077 [+0.065, +0.092]	yes	+0.038 [+0.031, +0.045]	yes
Transformer	FD002	CP-norm	+0.077 [+0.060, +0.093]	yes	+0.035 [+0.028, +0.041]	yes
Transformer	FD003	Heterosc.	+0.087 [+0.049, +0.121]	yes	+0.049 [+0.033, +0.066]	yes
Transformer	FD003	CP-norm	+0.083 [+0.041, +0.126]	yes	+0.066 [+0.052, +0.081]	yes
Transformer	FD004	Heterosc.	+0.108 [+0.093, +0.124]	yes	+0.041 [+0.025, +0.055]	yes
Transformer	FD004	CP-norm	+0.098 [+0.087, +0.111]	yes	+0.046 [+0.029, +0.060]	yes

S2. TERMINAL-INSTANT DECISION STRESS TEST, FULL RESULTS

Source: results/decision/maintenance_decision_one_sided.json. One-sided 95% lower bound; unrec. = unrecognised maintenance need (overall / conditional on true $RUL \leq L$); prem. = premature trigger rate; $RUL@t$ = true RUL at premature trigger under the 125-cycle label clip (unclipped values for the main-text rows are in Table VII of the paper); $cost = c_{late}r_{late} + c_{early}r_{early}$ with $c_{early} = 1$; rank at ratios 5/20/100.

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	$RUL@t$	cost 5/20/100	rank
LSTM	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	4/4/4
LSTM	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Clean	20	Heterosc.	0.0020 / 0.011	0.0000	–	0.010 / 0.040 / 0.200	3/5/5
LSTM	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0220	54.6	0.022 / 0.022 / 0.022	4/3/3
LSTM	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0220	54.6	0.022 / 0.022 / 0.022	5/4/4
LSTM	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0020	46.0	0.002 / 0.002 / 0.002	2/2/2
LSTM	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	60.5	0.008 / 0.008 / 0.008	2/2/2
LSTM	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	4/4/4
LSTM	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	5/5/5
LSTM	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0220	69.7	0.022 / 0.022 / 0.022	3/3/3
LSTM	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0172	36.7	0.017 / 0.017 / 0.017	4/4/4
LSTM	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0004	33.0	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Noise 1%	20	Heterosc.	0.0020 / 0.011	0.0000	–	0.010 / 0.040 / 0.200	3/5/5
LSTM	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	54.3	0.021 / 0.021 / 0.021	4/3/2
LSTM	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0232	55.8	0.023 / 0.023 / 0.023	5/4/3
LSTM	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	20	CP-norm	0.0004 / 0.002	0.0020	46.0	0.004 / 0.010 / 0.042	2/2/4
LSTM	FD001	Noise 1%	30	Heterosc.	0.0008 / 0.003	0.0072	62.2	0.011 / 0.023 / 0.087	2/3/5
LSTM	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0296	68.1	0.030 / 0.030 / 0.030	4/4/3
LSTM	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	5/5/4
LSTM	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0020	68.0	0.002 / 0.002 / 0.002	1/1/1
LSTM	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0188	65.8	0.019 / 0.019 / 0.019	3/2/2
LSTM	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0028	37.0	0.003 / 0.003 / 0.003	1/1/1
LSTM	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	38.4	0.019 / 0.019 / 0.019	4/4/4
LSTM	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0192	39.0	0.019 / 0.019 / 0.019	5/5/5
LSTM	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0040	37.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0056	39.5	0.006 / 0.006 / 0.006	3/3/3
LSTM	FD001	Bias 5%	20	Heterosc.	0.0052 / 0.027	0.0008	57.0	0.027 / 0.105 / 0.521	3/5/5
LSTM	FD001	Bias 5%	20	Fixed var.	0.0016 / 0.008	0.0264	59.2	0.034 / 0.058 / 0.186	5/3/3
LSTM	FD001	Bias 5%	20	MC Drop.	0.0012 / 0.006	0.0272	58.9	0.033 / 0.051 / 0.147	4/2/2
LSTM	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Bias 5%	20	CP-norm	0.0028 / 0.015	0.0120	75.1	0.026 / 0.068 / 0.292	2/4/4
LSTM	FD001	Bias 5%	30	Heterosc.	0.0032 / 0.013	0.0144	68.4	0.030 / 0.078 / 0.334	2/5/5
LSTM	FD001	Bias 5%	30	Fixed var.	0.0012 / 0.005	0.0316	69.1	0.038 / 0.056 / 0.152	3/2/2
LSTM	FD001	Bias 5%	30	MC Drop.	0.0012 / 0.005	0.0336	69.3	0.040 / 0.058 / 0.154	4/3/3
LSTM	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0220	68.1	0.022 / 0.022 / 0.022	1/1/1
LSTM	FD001	Bias 5%	30	CP-norm	0.0016 / 0.006	0.0384	77.3	0.046 / 0.070 / 0.198	5/4/4

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0180	35.2	0.018 / 0.018 / 0.018	1/1/1
LSTM	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0720	48.4	0.072 / 0.072 / 0.072	4/4/4
LSTM	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0740	49.4	0.074 / 0.074 / 0.074	5/5/5
LSTM	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	35.0	0.020 / 0.020 / 0.020	2/2/2
LSTM	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0280	38.3	0.028 / 0.028 / 0.028	3/3/3
LSTM	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0260	62.0	0.026 / 0.026 / 0.026	1/1/1
LSTM	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1180	61.5	0.118 / 0.118 / 0.118	4/4/4
LSTM	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1200	61.3	0.120 / 0.120 / 0.120	5/5/5
LSTM	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0300	54.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0580	66.8	0.058 / 0.058 / 0.058	3/3/3
LSTM	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0680	68.3	0.068 / 0.068 / 0.068	1/1/1
LSTM	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1360	77.8	0.136 / 0.136 / 0.136	3/3/3
LSTM	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1400	77.6	0.140 / 0.140 / 0.140	4/4/4
LSTM	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1600	78.1	0.160 / 0.160 / 0.160	5/5/5
LSTM	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1340	80.9	0.134 / 0.134 / 0.134	2/2/2
LSTM	FD002	Clean	10	Heterosc.	0.0000 / 0.000	0.0008	36.0	0.001 / 0.001 / 0.001	1/1/1
LSTM	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0617	42.4	0.062 / 0.062 / 0.062	4/4/4
LSTM	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0625	42.4	0.062 / 0.062 / 0.062	5/5/5
LSTM	FD002	Clean	10	Ensemble	0.0000 / 0.000	0.0040	42.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD002	Clean	10	CP-norm	0.0000 / 0.000	0.0095	48.8	0.009 / 0.009 / 0.009	3/3/3
LSTM	FD002	Clean	20	Heterosc.	0.0000 / 0.000	0.0372	49.4	0.037 / 0.037 / 0.037	1/1/1
LSTM	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0783	52.6	0.078 / 0.078 / 0.078	4/4/4
LSTM	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0814	53.0	0.081 / 0.081 / 0.081	5/5/5
LSTM	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0514	48.2	0.051 / 0.051 / 0.051	2/2/2
LSTM	FD002	Clean	20	CP-norm	0.0000 / 0.000	0.0522	54.3	0.052 / 0.052 / 0.052	3/3/3
LSTM	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0743	68.0	0.074 / 0.074 / 0.074	3/3/3
LSTM	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0719	64.0	0.072 / 0.072 / 0.072	1/1/1
LSTM	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0719	64.2	0.072 / 0.072 / 0.072	2/2/2
LSTM	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0870	67.1	0.087 / 0.087 / 0.087	5/5/5
LSTM	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0838	69.8	0.084 / 0.084 / 0.084	4/4/4
LSTM	FD002	Noise 1%	10	Heterosc.	0.0032 / 0.029	0.0068	44.3	0.023 / 0.070 / 0.323	1/2/4
LSTM	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0757	46.3	0.076 / 0.076 / 0.076	4/3/1
LSTM	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0778	46.8	0.078 / 0.078 / 0.078	5/5/2
LSTM	FD002	Noise 1%	10	Ensemble	0.0032 / 0.029	0.0126	39.8	0.028 / 0.076 / 0.329	2/4/5
LSTM	FD002	Noise 1%	10	CP-norm	0.0027 / 0.024	0.0150	50.7	0.028 / 0.069 / 0.284	3/1/3
LSTM	FD002	Noise 1%	20	Heterosc.	0.0022 / 0.010	0.0496	56.1	0.061 / 0.094 / 0.271	1/2/5
LSTM	FD002	Noise 1%	20	Fixed var.	0.0003 / 0.001	0.0877	56.8	0.089 / 0.094 / 0.119	4/3/1
LSTM	FD002	Noise 1%	20	MC Drop.	0.0003 / 0.001	0.0889	57.0	0.090 / 0.095 / 0.120	5/4/2
LSTM	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0704	57.3	0.074 / 0.086 / 0.149	3/1/3
LSTM	FD002	Noise 1%	20	CP-norm	0.0016 / 0.007	0.0653	61.8	0.073 / 0.097 / 0.223	2/5/4
LSTM	FD002	Noise 1%	30	Heterosc.	0.0003 / 0.001	0.0938	77.8	0.095 / 0.100 / 0.125	3/3/5
LSTM	FD002	Noise 1%	30	Fixed var.	0.0002 / 0.001	0.0822	71.4	0.083 / 0.085 / 0.098	1/2/2
LSTM	FD002	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0841	71.8	0.084 / 0.084 / 0.084	2/1/1
LSTM	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.1075	77.2	0.108 / 0.108 / 0.108	5/5/3
LSTM	FD002	Noise 1%	30	CP-norm	0.0002 / 0.001	0.1018	78.5	0.103 / 0.105 / 0.118	4/4/4
LSTM	FD002	Bias 5%	10	Heterosc.	0.0553 / 0.500	0.1711	89.2	0.448 / 1.278 / 5.705	5/5/5
LSTM	FD002	Bias 5%	10	Fixed var.	0.0370 / 0.334	0.1521	90.5	0.337 / 0.892 / 3.852	2/2/2
LSTM	FD002	Bias 5%	10	MC Drop.	0.0360 / 0.326	0.1521	90.5	0.332 / 0.873 / 3.757	1/1/1
LSTM	FD002	Bias 5%	10	Ensemble	0.0395 / 0.357	0.1684	86.6	0.366 / 0.959 / 4.121	3/3/3
LSTM	FD002	Bias 5%	10	CP-norm	0.0536 / 0.484	0.1700	88.6	0.438 / 1.242 / 5.530	4/4/4
LSTM	FD002	Bias 5%	20	Heterosc.	0.0783 / 0.367	0.1698	94.3	0.561 / 1.735 / 7.996	5/5/5
LSTM	FD002	Bias 5%	20	Fixed var.	0.0721 / 0.338	0.1410	95.0	0.502 / 1.583 / 7.351	3/3/4
LSTM	FD002	Bias 5%	20	MC Drop.	0.0705 / 0.330	0.1412	94.9	0.494 / 1.551 / 7.193	2/2/2
LSTM	FD002	Bias 5%	20	Ensemble	0.0451 / 0.211	0.1905	92.9	0.416 / 1.092 / 4.696	1/1/1
LSTM	FD002	Bias 5%	20	CP-norm	0.0707 / 0.331	0.1698	93.9	0.523 / 1.583 / 7.237	4/4/3
LSTM	FD002	Bias 5%	30	Heterosc.	0.0590 / 0.241	0.1609	99.1	0.456 / 1.340 / 6.058	3/3/3
LSTM	FD002	Bias 5%	30	Fixed var.	0.0751 / 0.306	0.1300	99.2	0.505 / 1.632 / 7.640	5/5/5
LSTM	FD002	Bias 5%	30	MC Drop.	0.0729 / 0.297	0.1303	99.1	0.495 / 1.588 / 7.419	4/4/4
LSTM	FD002	Bias 5%	30	Ensemble	0.0348 / 0.142	0.1897	98.1	0.364 / 0.885 / 3.668	1/1/1
LSTM	FD002	Bias 5%	30	CP-norm	0.0582 / 0.237	0.1606	98.8	0.452 / 1.324 / 5.979	2/2/2
LSTM	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.7154	89.4	0.715 / 0.715 / 0.715	2/2/2
LSTM	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7518	90.8	0.752 / 0.752 / 0.752	4/4/4
LSTM	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7518	90.8	0.752 / 0.752 / 0.752	5/5/5
LSTM	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7431	90.5	0.743 / 0.743 / 0.743	3/3/3
LSTM	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7115	89.3	0.711 / 0.711 / 0.711	1/1/1
LSTM	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6933	95.1	0.693 / 0.693 / 0.693	2/2/2
LSTM	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	4/4/4
LSTM	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	5/5/5
LSTM	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6957	95.2	0.696 / 0.696 / 0.696	3/3/3
LSTM	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6917	95.0	0.692 / 0.692 / 0.692	1/1/1
LSTM	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6435	99.6	0.643 / 0.643 / 0.643	1/1/1
LSTM	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	3/3/3
LSTM	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	4/4/4
LSTM	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	5/5/5
LSTM	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6435	99.6	0.643 / 0.643 / 0.643	2/2/2

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0160	48.1	0.016 / 0.016 / 0.016	5/5/5
LSTM	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0140	45.0	0.014 / 0.014 / 0.014	4/4/4
LSTM	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0380	54.5	0.038 / 0.038 / 0.038	4/4/4
LSTM	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0420	54.0	0.042 / 0.042 / 0.042	5/5/5
LSTM	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0320	63.6	0.032 / 0.032 / 0.032	3/3/3
LSTM	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0140	60.9	0.014 / 0.014 / 0.014	1/1/1
LSTM	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0440	59.5	0.044 / 0.044 / 0.044	3/3/3
LSTM	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0500	59.1	0.050 / 0.050 / 0.050	4/4/4
LSTM	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	59.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0680	64.4	0.068 / 0.068 / 0.068	5/5/5
LSTM	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0148	46.4	0.015 / 0.015 / 0.015	4/4/4
LSTM	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0148	46.4	0.015 / 0.015 / 0.015	5/5/5
LSTM	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0012	68.0	0.001 / 0.001 / 0.001	3/3/3
LSTM	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0008	54.0	0.001 / 0.001 / 0.001	2/2/2
LSTM	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0340	54.7	0.034 / 0.034 / 0.034	4/4/4
LSTM	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0352	54.3	0.035 / 0.035 / 0.035	5/5/5
LSTM	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0292	63.5	0.029 / 0.029 / 0.029	3/3/3
LSTM	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0160	60.8	0.016 / 0.016 / 0.016	1/1/1
LSTM	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0496	59.3	0.050 / 0.050 / 0.050	3/3/3
LSTM	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0536	59.0	0.054 / 0.054 / 0.054	4/4/4
LSTM	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	58.7	0.034 / 0.034 / 0.034	2/2/2
LSTM	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0740	65.1	0.074 / 0.074 / 0.074	5/5/5
LSTM	FD003	Bias 5%	10	Heterosc.	0.0016 / 0.020	0.0024	36.7	0.010 / 0.034 / 0.162	3/3/5
LSTM	FD003	Bias 5%	10	Fixed var.	0.0008 / 0.010	0.0300	46.4	0.034 / 0.046 / 0.110	4/4/3
LSTM	FD003	Bias 5%	10	MC Drop.	0.0008 / 0.010	0.0328	47.5	0.037 / 0.049 / 0.113	5/5/4
LSTM	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	34.0	0.002 / 0.002 / 0.002	1/1/1
LSTM	FD003	Bias 5%	10	CP-norm	0.0004 / 0.005	0.0048	49.2	0.007 / 0.013 / 0.045	2/2/2
LSTM	FD003	Bias 5%	20	Heterosc.	0.0008 / 0.006	0.0060	55.2	0.010 / 0.022 / 0.086	2/2/3
LSTM	FD003	Bias 5%	20	Fixed var.	0.0008 / 0.006	0.0572	54.7	0.061 / 0.073 / 0.137	4/4/4
LSTM	FD003	Bias 5%	20	MC Drop.	0.0008 / 0.006	0.0592	54.8	0.063 / 0.075 / 0.139	5/5/5
LSTM	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	57.3	0.006 / 0.006 / 0.006	1/1/1
LSTM	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0472	63.2	0.047 / 0.047 / 0.047	3/3/2
LSTM	FD003	Bias 5%	30	Heterosc.	0.0012 / 0.006	0.0384	59.5	0.044 / 0.062 / 0.158	1/2/4
LSTM	FD003	Bias 5%	30	Fixed var.	0.0008 / 0.004	0.0696	62.5	0.074 / 0.086 / 0.150	3/4/3
LSTM	FD003	Bias 5%	30	MC Drop.	0.0004 / 0.002	0.0728	62.4	0.075 / 0.081 / 0.113	4/3/2
LSTM	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0620	60.7	0.062 / 0.062 / 0.062	2/1/1
LSTM	FD003	Bias 5%	30	CP-norm	0.0020 / 0.010	0.1004	67.9	0.110 / 0.140 / 0.300	5/5/5
LSTM	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0220	47.4	0.022 / 0.022 / 0.022	2/2/2
LSTM	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2360	56.2	0.236 / 0.236 / 0.236	4/4/4
LSTM	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.2400	56.3	0.240 / 0.240 / 0.240	5/5/5
LSTM	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	52.0	0.020 / 0.020 / 0.020	1/1/1
LSTM	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0440	46.1	0.044 / 0.044 / 0.044	3/3/3
LSTM	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1380	55.1	0.138 / 0.138 / 0.138	1/1/1
LSTM	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3540	68.7	0.354 / 0.354 / 0.354	4/4/4
LSTM	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3600	69.1	0.360 / 0.360 / 0.360	5/5/5
LSTM	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.1600	55.9	0.160 / 0.160 / 0.160	2/2/2
LSTM	FD003	Drift 5%	20	CP-norm	0.0000 / 0.000	0.1880	60.7	0.188 / 0.188 / 0.188	3/3/3
LSTM	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.2500	71.3	0.250 / 0.250 / 0.250	1/1/1
LSTM	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4220	82.6	0.422 / 0.422 / 0.422	4/4/4
LSTM	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4280	82.7	0.428 / 0.428 / 0.428	5/5/5
LSTM	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.2600	70.5	0.260 / 0.260 / 0.260	2/2/2
LSTM	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3540	80.9	0.354 / 0.354 / 0.354	3/3/3
LSTM	FD004	Clean	10	Heterosc.	0.0034 / 0.044	0.0329	42.6	0.050 / 0.100 / 0.370	1/5/5
LSTM	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0726	42.7	0.073 / 0.073 / 0.073	4/3/3
LSTM	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0734	42.8	0.073 / 0.073 / 0.073	5/4/4
LSTM	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0506	43.8	0.051 / 0.051 / 0.051	2/1/1
LSTM	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0650	47.9	0.065 / 0.065 / 0.065	3/2/2
LSTM	FD004	Clean	20	Heterosc.	0.0008 / 0.005	0.0388	60.0	0.043 / 0.056 / 0.123	1/2/3
LSTM	FD004	Clean	20	Fixed var.	0.0017 / 0.010	0.0363	57.1	0.045 / 0.070 / 0.205	2/4/4
LSTM	FD004	Clean	20	MC Drop.	0.0017 / 0.010	0.0388	56.9	0.047 / 0.073 / 0.208	4/5/5
LSTM	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0464	60.5	0.046 / 0.046 / 0.046	3/1/1
LSTM	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0675	69.7	0.068 / 0.068 / 0.068	5/3/2
LSTM	FD004	Clean	30	Heterosc.	0.0025 / 0.011	0.0684	77.6	0.081 / 0.119 / 0.322	3/1/2
LSTM	FD004	Clean	30	Fixed var.	0.0042 / 0.019	0.0371	72.6	0.058 / 0.122 / 0.459	1/2/3
LSTM	FD004	Clean	30	MC Drop.	0.0042 / 0.019	0.0380	72.4	0.059 / 0.122 / 0.460	2/3/4
LSTM	FD004	Clean	30	Ensemble	0.0042 / 0.019	0.0675	75.0	0.089 / 0.152 / 0.489	4/5/5
LSTM	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.1055	80.7	0.110 / 0.122 / 0.190	5/4/1

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD004	Noise 1%	10	Heterosc.	0.0056 / 0.073	0.0441	47.9	0.072 / 0.155 / 0.601	2/5/5
LSTM	FD004	Noise 1%	10	Fixed var.	0.0010 / 0.013	0.0734	45.0	0.078 / 0.094 / 0.175	3/2/2
LSTM	FD004	Noise 1%	10	MC Drop.	0.0008 / 0.011	0.0751	45.3	0.079 / 0.092 / 0.159	4/1/1
LSTM	FD004	Noise 1%	10	Ensemble	0.0025 / 0.033	0.0523	46.3	0.065 / 0.103 / 0.305	1/3/3
LSTM	FD004	Noise 1%	10	CP-norm	0.0029 / 0.038	0.0759	53.6	0.090 / 0.133 / 0.363	5/4/4
LSTM	FD004	Noise 1%	20	Heterosc.	0.0025 / 0.015	0.0550	67.2	0.068 / 0.106 / 0.308	3/4/5
LSTM	FD004	Noise 1%	20	Fixed var.	0.0025 / 0.015	0.0523	62.2	0.065 / 0.103 / 0.305	2/3/4
LSTM	FD004	Noise 1%	20	MC Drop.	0.0022 / 0.013	0.0538	62.1	0.065 / 0.098 / 0.273	1/2/3
LSTM	FD004	Noise 1%	20	Ensemble	0.0017 / 0.010	0.0616	66.0	0.070 / 0.095 / 0.230	4/1/2
LSTM	FD004	Noise 1%	20	CP-norm	0.0014 / 0.008	0.0906	73.8	0.097 / 0.118 / 0.226	5/5/1
LSTM	FD004	Noise 1%	30	Heterosc.	0.0027 / 0.012	0.0857	81.4	0.099 / 0.140 / 0.356	3/1/3
LSTM	FD004	Noise 1%	30	Fixed var.	0.0054 / 0.024	0.0557	77.7	0.083 / 0.164 / 0.596	1/5/5
LSTM	FD004	Noise 1%	30	MC Drop.	0.0052 / 0.023	0.0582	78.2	0.084 / 0.163 / 0.581	2/4/4
LSTM	FD004	Noise 1%	30	Ensemble	0.0025 / 0.011	0.0928	78.9	0.105 / 0.143 / 0.346	4/2/2
LSTM	FD004	Noise 1%	30	CP-norm	0.0014 / 0.006	0.1200	83.9	0.127 / 0.147 / 0.255	5/3/1
LSTM	FD004	Bias 5%	10	Heterosc.	0.0164 / 0.216	0.2889	83.7	0.371 / 0.616 / 1.926	1/3/5
LSTM	FD004	Bias 5%	10	Fixed var.	0.0140 / 0.184	0.3507	86.2	0.421 / 0.631 / 1.752	4/5/3
LSTM	FD004	Bias 5%	10	MC Drop.	0.0138 / 0.182	0.3529	86.2	0.422 / 0.630 / 1.737	5/4/2
LSTM	FD004	Bias 5%	10	Ensemble	0.0110 / 0.144	0.3367	86.7	0.392 / 0.556 / 1.434	3/1/1
LSTM	FD004	Bias 5%	10	CP-norm	0.0145 / 0.191	0.3151	84.4	0.388 / 0.605 / 1.767	2/2/4
LSTM	FD004	Bias 5%	20	Heterosc.	0.0255 / 0.151	0.3016	94.5	0.429 / 0.811 / 2.850	2/3/3
LSTM	FD004	Bias 5%	20	Fixed var.	0.0257 / 0.152	0.3411	95.1	0.469 / 0.854 / 2.906	4/5/5
LSTM	FD004	Bias 5%	20	MC Drop.	0.0255 / 0.151	0.3431	95.1	0.471 / 0.853 / 2.892	5/4/4
LSTM	FD004	Bias 5%	20	Ensemble	0.0135 / 0.080	0.3468	96.2	0.414 / 0.617 / 1.697	1/1/1
LSTM	FD004	Bias 5%	20	CP-norm	0.0224 / 0.133	0.3219	94.7	0.434 / 0.771 / 2.567	3/2/2
LSTM	FD004	Bias 5%	30	Heterosc.	0.0300 / 0.132	0.3229	99.3	0.473 / 0.924 / 3.327	2/3/3
LSTM	FD004	Bias 5%	30	Fixed var.	0.0322 / 0.141	0.3389	99.4	0.500 / 0.984 / 3.563	5/5/5
LSTM	FD004	Bias 5%	30	MC Drop.	0.0305 / 0.134	0.3414	99.5	0.494 / 0.952 / 3.396	4/4/4
LSTM	FD004	Bias 5%	30	Ensemble	0.0186 / 0.081	0.3519	100.2	0.445 / 0.723 / 2.208	1/1/1
LSTM	FD004	Bias 5%	30	CP-norm	0.0278 / 0.122	0.3399	99.3	0.479 / 0.897 / 3.125	3/2/2
LSTM	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.4658	81.8	0.466 / 0.466 / 0.466	1/1/1
LSTM	FD004	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.6329	87.3	0.633 / 0.633 / 0.633	3/3/3
LSTM	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.6397	87.6	0.640 / 0.640 / 0.640	5/5/5
LSTM	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.6371	88.3	0.637 / 0.637 / 0.637	4/4/4
LSTM	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.5882	86.9	0.588 / 0.588 / 0.588	2/2/2
LSTM	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.5899	96.0	0.590 / 0.590 / 0.590	1/1/1
LSTM	FD004	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6346	97.2	0.635 / 0.635 / 0.635	2/2/2
LSTM	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6346	97.2	0.635 / 0.635 / 0.635	3/3/3
LSTM	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6540	97.6	0.654 / 0.654 / 0.654	5/5/5
LSTM	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6422	97.5	0.642 / 0.642 / 0.642	4/4/4
LSTM	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6194	101.2	0.619 / 0.619 / 0.619	1/1/1
LSTM	FD004	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6253	101.3	0.625 / 0.625 / 0.625	2/2/2
LSTM	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6278	101.3	0.628 / 0.628 / 0.628	3/3/3
LSTM	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6287	101.3	0.629 / 0.629 / 0.629	4/4/4
LSTM	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6304	101.5	0.630 / 0.630 / 0.630	5/5/5
Transformer	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0220	36.0	0.022 / 0.022 / 0.022	5/5/5
Transformer	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0200	36.3	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0040	35.0	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0200	49.6	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0260	49.3	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	53.0	0.008 / 0.008 / 0.008	1/1/1
Transformer	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0220	67.0	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0240	67.1	0.024 / 0.024 / 0.024	4/4/4
Transformer	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0100	53.0	0.010 / 0.010 / 0.010	2/2/2
Transformer	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0380	68.8	0.038 / 0.038 / 0.038	5/5/5
Transformer	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0212	36.1	0.021 / 0.021 / 0.021	5/5/5
Transformer	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0208	36.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0040	35.4	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	49.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0260	50.9	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0084	53.7	0.008 / 0.008 / 0.008	2/2/2
Transformer	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0224	67.6	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0244	67.9	0.024 / 0.024 / 0.024	4/4/4
Transformer	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0060	53.0	0.006 / 0.006 / 0.006	1/1/1
Transformer	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0376	69.0	0.038 / 0.038 / 0.038	5/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0024	36.3	0.002 / 0.002 / 0.002	2/2/2
Transformer	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	36.4	0.019 / 0.019 / 0.019	4/4/4
Transformer	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0204	38.1	0.020 / 0.020 / 0.020	5/5/5
Transformer	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	37.0	0.002 / 0.002 / 0.002	1/1/1
Transformer	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0048	36.0	0.005 / 0.005 / 0.005	3/3/3
Transformer	FD001	Bias 5%	20	Heterosc.	0.0016 / 0.008	0.0000	–	0.008 / 0.032 / 0.160	3/5/5
Transformer	FD001	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0208	53.2	0.021 / 0.021 / 0.021	4/3/3
Transformer	FD001	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0236	53.2	0.024 / 0.024 / 0.024	5/4/4
Transformer	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0008	64.5	0.001 / 0.001 / 0.001	2/2/2
Transformer	FD001	Bias 5%	30	Heterosc.	0.0004 / 0.002	0.0104	60.8	0.012 / 0.018 / 0.050	2/2/5
Transformer	FD001	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0236	64.2	0.024 / 0.024 / 0.024	3/3/2
Transformer	FD001	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0260	64.7	0.026 / 0.026 / 0.026	4/4/3
Transformer	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0100	59.0	0.010 / 0.010 / 0.010	1/1/1
Transformer	FD001	Bias 5%	30	CP-norm	0.0000 / 0.000	0.0316	69.5	0.032 / 0.032 / 0.032	5/5/4
Transformer	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0160	35.9	0.016 / 0.016 / 0.016	1/1/1
Transformer	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0680	47.0	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0700	47.0	0.070 / 0.070 / 0.070	5/5/5
Transformer	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	35.0	0.020 / 0.020 / 0.020	3/3/3
Transformer	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0160	35.9	0.016 / 0.016 / 0.016	2/2/2
Transformer	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0240	54.5	0.024 / 0.024 / 0.024	2/2/2
Transformer	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1020	58.5	0.102 / 0.102 / 0.102	4/4/4
Transformer	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1040	59.1	0.104 / 0.104 / 0.104	5/5/5
Transformer	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0200	57.0	0.020 / 0.020 / 0.020	1/1/1
Transformer	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0360	64.1	0.036 / 0.036 / 0.036	3/3/3
Transformer	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0620	72.4	0.062 / 0.062 / 0.062	1/1/1
Transformer	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1320	75.3	0.132 / 0.132 / 0.132	4/4/4
Transformer	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1420	75.8	0.142 / 0.142 / 0.142	5/5/5
Transformer	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1300	72.8	0.130 / 0.130 / 0.130	2/2/2
Transformer	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1300	83.1	0.130 / 0.130 / 0.130	3/3/3
Transformer	FD002	Clean	10	Heterosc.	0.0032 / 0.029	0.0000	–	0.016 / 0.063 / 0.316	1/3/3
Transformer	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0427	40.6	0.043 / 0.043 / 0.043	5/2/2
Transformer	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0411	40.3	0.041 / 0.041 / 0.041	4/1/1
Transformer	FD002	Clean	10	Ensemble	0.0040 / 0.036	0.0000	–	0.020 / 0.079 / 0.395	2/4/4
Transformer	FD002	Clean	10	CP-norm	0.0040 / 0.036	0.0055	44.0	0.025 / 0.085 / 0.401	3/5/5
Transformer	FD002	Clean	20	Heterosc.	0.0008 / 0.004	0.0150	46.7	0.019 / 0.031 / 0.094	1/2/4
Transformer	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0625	51.9	0.062 / 0.062 / 0.062	4/4/2
Transformer	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0648	51.9	0.065 / 0.065 / 0.065	5/5/3
Transformer	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0198	44.2	0.020 / 0.020 / 0.020	2/1/1
Transformer	FD002	Clean	20	CP-norm	0.0008 / 0.004	0.0316	52.9	0.036 / 0.047 / 0.111	3/3/5
Transformer	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0601	69.9	0.060 / 0.060 / 0.060	1/1/1
Transformer	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0632	63.7	0.063 / 0.063 / 0.063	2/2/2
Transformer	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0680	64.1	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0711	65.3	0.071 / 0.071 / 0.071	5/5/5
Transformer	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0664	70.1	0.066 / 0.066 / 0.066	3/3/3
Transformer	FD002	Noise 1%	10	Heterosc.	0.0028 / 0.026	0.0051	38.0	0.019 / 0.062 / 0.290	2/3/4
Transformer	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0667	47.5	0.067 / 0.067 / 0.067	5/4/2
Transformer	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0610	46.3	0.061 / 0.061 / 0.061	4/2/1
Transformer	FD002	Noise 1%	10	Ensemble	0.0008 / 0.007	0.0079	42.0	0.012 / 0.024 / 0.087	1/1/3
Transformer	FD002	Noise 1%	10	CP-norm	0.0030 / 0.027	0.0115	43.9	0.027 / 0.072 / 0.312	3/5/5
Transformer	FD002	Noise 1%	20	Heterosc.	0.0021 / 0.010	0.0308	56.2	0.041 / 0.072 / 0.236	1/2/4
Transformer	FD002	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0817	60.2	0.082 / 0.082 / 0.082	5/4/2
Transformer	FD002	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0816	59.8	0.082 / 0.082 / 0.082	4/3/1
Transformer	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0395	55.8	0.043 / 0.055 / 0.119	2/1/3
Transformer	FD002	Noise 1%	20	CP-norm	0.0021 / 0.010	0.0474	62.2	0.058 / 0.089 / 0.253	3/5/5
Transformer	FD002	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0813	77.3	0.081 / 0.081 / 0.081	1/1/1
Transformer	FD002	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0889	73.2	0.089 / 0.089 / 0.089	2/2/2
Transformer	FD002	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0920	73.5	0.092 / 0.092 / 0.092	4/4/4
Transformer	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0909	75.2	0.091 / 0.091 / 0.091	3/3/3
Transformer	FD002	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0963	78.7	0.096 / 0.096 / 0.096	5/5/5
Transformer	FD002	Bias 5%	10	Heterosc.	0.0655 / 0.591	0.1393	88.7	0.467 / 1.448 / 6.685	4/5/5
Transformer	FD002	Bias 5%	10	Fixed var.	0.0411 / 0.371	0.1507	90.1	0.356 / 0.973 / 4.261	1/1/1
Transformer	FD002	Bias 5%	10	MC Drop.	0.0443 / 0.400	0.1499	90.2	0.371 / 1.035 / 4.577	2/2/2
Transformer	FD002	Bias 5%	10	Ensemble	0.0530 / 0.479	0.1502	90.8	0.415 / 1.209 / 5.447	3/3/3
Transformer	FD002	Bias 5%	10	CP-norm	0.0643 / 0.581	0.1458	87.8	0.468 / 1.433 / 6.581	5/4/4
Transformer	FD002	Bias 5%	20	Heterosc.	0.1021 / 0.479	0.1380	94.3	0.649 / 2.181 / 10.351	5/5/5
Transformer	FD002	Bias 5%	20	Fixed var.	0.0759 / 0.356	0.1432	94.0	0.523 / 1.661 / 7.732	2/2/2
Transformer	FD002	Bias 5%	20	MC Drop.	0.0794 / 0.372	0.1421	94.4	0.539 / 1.729 / 8.079	3/3/3
Transformer	FD002	Bias 5%	20	Ensemble	0.0585 / 0.274	0.1447	93.9	0.437 / 1.315 / 5.994	1/1/1
Transformer	FD002	Bias 5%	20	CP-norm	0.0919 / 0.430	0.1428	93.4	0.602 / 1.980 / 9.329	4/4/4
Transformer	FD002	Bias 5%	30	Heterosc.	0.0821 / 0.335	0.1334	98.4	0.544 / 1.775 / 8.339	5/5/5
Transformer	FD002	Bias 5%	30	Fixed var.	0.0776 / 0.317	0.1345	97.9	0.523 / 1.687 / 7.897	4/4/4
Transformer	FD002	Bias 5%	30	MC Drop.	0.0773 / 0.315	0.1344	97.9	0.521 / 1.681 / 7.866	3/3/3
Transformer	FD002	Bias 5%	30	Ensemble	0.0530 / 0.216	0.1407	96.8	0.406 / 1.200 / 5.437	1/1/1
Transformer	FD002	Bias 5%	30	CP-norm	0.0729 / 0.297	0.1363	98.1	0.501 / 1.594 / 7.425	2/2/2

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.6909	88.4	0.691 / 0.691 / 0.691	1/1/1
Transformer	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7502	90.8	0.750 / 0.750 / 0.750	5/5/5
Transformer	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7494	90.7	0.749 / 0.749 / 0.749	4/4/4
Transformer	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7154	89.4	0.715 / 0.715 / 0.715	3/3/3
Transformer	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7043	89.0	0.704 / 0.704 / 0.704	2/2/2
Transformer	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6917	95.0	0.692 / 0.692 / 0.692	1/1/1
Transformer	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6988	95.3	0.699 / 0.699 / 0.699	3/3/3
Transformer	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6988	95.3	0.699 / 0.699 / 0.699	4/4/4
Transformer	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	5/5/5
Transformer	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6925	95.0	0.692 / 0.692 / 0.692	2/2/2
Transformer	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6427	99.5	0.643 / 0.643 / 0.643	1/1/1
Transformer	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	3/3/3
Transformer	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	4/4/4
Transformer	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	5/5/5
Transformer	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6427	99.5	0.643 / 0.643 / 0.643	2/2/2
Transformer	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	4/4/4
Transformer	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	5/5/5
Transformer	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0060	51.0	0.006 / 0.006 / 0.006	3/3/3
Transformer	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0180	48.3	0.018 / 0.018 / 0.018	5/5/5
Transformer	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0140	49.0	0.014 / 0.014 / 0.014	3/3/3
Transformer	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0160	68.1	0.016 / 0.016 / 0.016	4/4/4
Transformer	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0220	57.2	0.022 / 0.022 / 0.022	1/1/1
Transformer	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0320	57.4	0.032 / 0.032 / 0.032	3/3/3
Transformer	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0320	57.4	0.032 / 0.032 / 0.032	4/4/4
Transformer	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	59.3	0.030 / 0.030 / 0.030	2/2/2
Transformer	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0700	62.2	0.070 / 0.070 / 0.070	5/5/5
Transformer	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	4/4/4
Transformer	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	5/5/5
Transformer	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0060	52.7	0.006 / 0.006 / 0.006	3/3/3
Transformer	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0004	54.0	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0156	49.0	0.016 / 0.016 / 0.016	5/5/5
Transformer	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0128	50.5	0.013 / 0.013 / 0.013	3/3/3
Transformer	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0148	66.5	0.015 / 0.015 / 0.015	4/4/4
Transformer	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0224	59.3	0.022 / 0.022 / 0.022	1/1/1
Transformer	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0332	57.2	0.033 / 0.033 / 0.033	2/2/2
Transformer	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0332	57.2	0.033 / 0.033 / 0.033	3/3/3
Transformer	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	58.7	0.034 / 0.034 / 0.034	4/4/4
Transformer	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0676	62.7	0.068 / 0.068 / 0.068	5/5/5
Transformer	FD003	Bias 5%	10	Heterosc.	0.0004 / 0.005	0.0020	34.0	0.004 / 0.010 / 0.042	2/2/4
Transformer	FD003	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0132	37.3	0.013 / 0.013 / 0.013	5/4/3
Transformer	FD003	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0128	37.3	0.013 / 0.013 / 0.013	3/3/2
Transformer	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Bias 5%	10	CP-norm	0.0008 / 0.010	0.0088	52.6	0.013 / 0.025 / 0.089	4/5/5
Transformer	FD003	Bias 5%	20	Heterosc.	0.0000 / 0.000	0.0052	61.6	0.005 / 0.005 / 0.005	1/1/1
Transformer	FD003	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0396	49.8	0.040 / 0.040 / 0.040	5/5/5
Transformer	FD003	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0348	49.7	0.035 / 0.035 / 0.035	4/4/4
Transformer	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	52.0	0.006 / 0.006 / 0.006	2/2/2
Transformer	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0172	65.3	0.017 / 0.017 / 0.017	3/3/3
Transformer	FD003	Bias 5%	30	Heterosc.	0.0000 / 0.000	0.0380	60.2	0.038 / 0.038 / 0.038	1/1/1
Transformer	FD003	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0456	58.0	0.046 / 0.046 / 0.046	3/3/3
Transformer	FD003	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0444	57.8	0.044 / 0.044 / 0.044	2/2/2
Transformer	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0600	57.9	0.060 / 0.060 / 0.060	4/4/4
Transformer	FD003	Bias 5%	30	CP-norm	0.0004 / 0.002	0.0764	65.9	0.078 / 0.084 / 0.116	5/5/5
Transformer	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0460	47.3	0.046 / 0.046 / 0.046	1/1/1
Transformer	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2040	52.4	0.204 / 0.204 / 0.204	5/5/5
Transformer	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.1920	51.4	0.192 / 0.192 / 0.192	4/4/4
Transformer	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0500	46.0	0.050 / 0.050 / 0.050	2/2/2
Transformer	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0820	49.2	0.082 / 0.082 / 0.082	3/3/3
Transformer	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1740	57.0	0.174 / 0.174 / 0.174	1/1/1
Transformer	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3480	69.9	0.348 / 0.348 / 0.348	5/5/5
Transformer	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3260	67.6	0.326 / 0.326 / 0.326	4/4/4
Transformer	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.2000	58.2	0.200 / 0.200 / 0.200	3/3/3
Transformer	FD003	Drift 5%	20	CP-norm	0.0000 / 0.000	0.1980	60.2	0.198 / 0.198 / 0.198	2/2/2
Transformer	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.3060	76.0	0.306 / 0.306 / 0.306	1/1/1
Transformer	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4380	85.0	0.438 / 0.438 / 0.438	4/4/4
Transformer	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4440	84.9	0.444 / 0.444 / 0.444	5/5/5
Transformer	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.4100	81.7	0.410 / 0.410 / 0.410	3/3/3
Transformer	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3420	79.6	0.342 / 0.342 / 0.342	2/2/2

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD004	Clean	10	Heterosc.	0.0000 / 0.000	0.0304	42.7	0.030 / 0.030 / 0.030	1/1/1
Transformer	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0540	40.5	0.054 / 0.054 / 0.054	4/4/4
Transformer	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0430	40.4	0.043 / 0.043 / 0.043	2/2/2
Transformer	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0464	44.5	0.046 / 0.046 / 0.046	3/3/3
Transformer	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0557	44.1	0.056 / 0.056 / 0.056	5/5/5
Transformer	FD004	Clean	20	Heterosc.	0.0000 / 0.000	0.0287	64.1	0.029 / 0.029 / 0.029	1/1/1
Transformer	FD004	Clean	20	Fixed var.	0.0000 / 0.000	0.0405	58.0	0.041 / 0.041 / 0.041	4/4/4
Transformer	FD004	Clean	20	MC Drop.	0.0000 / 0.000	0.0312	56.7	0.031 / 0.031 / 0.031	2/2/2
Transformer	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0338	62.8	0.034 / 0.034 / 0.034	3/3/3
Transformer	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0565	66.0	0.057 / 0.057 / 0.057	5/5/5
Transformer	FD004	Clean	30	Heterosc.	0.0000 / 0.000	0.0473	75.2	0.047 / 0.047 / 0.047	4/2/2
Transformer	FD004	Clean	30	Fixed var.	0.0008 / 0.004	0.0422	70.5	0.046 / 0.059 / 0.127	2/4/4
Transformer	FD004	Clean	30	MC Drop.	0.0008 / 0.004	0.0371	70.5	0.041 / 0.054 / 0.122	1/3/3
Transformer	FD004	Clean	30	Ensemble	0.0000 / 0.000	0.0464	71.3	0.046 / 0.046 / 0.046	3/1/1
Transformer	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.0810	78.6	0.085 / 0.098 / 0.165	5/5/5
Transformer	FD004	Noise 1%	10	Heterosc.	0.0008 / 0.011	0.0378	45.4	0.042 / 0.055 / 0.122	1/2/5
Transformer	FD004	Noise 1%	10	Fixed var.	0.0003 / 0.004	0.0700	45.2	0.072 / 0.077 / 0.104	5/5/4
Transformer	FD004	Noise 1%	10	MC Drop.	0.0003 / 0.004	0.0522	42.3	0.054 / 0.066 / 0.086	3/3/2
Transformer	FD004	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0489	45.1	0.049 / 0.049 / 0.049	2/1/1
Transformer	FD004	Noise 1%	10	CP-norm	0.0003 / 0.004	0.0638	50.5	0.065 / 0.071 / 0.098	4/4/3
Transformer	FD004	Noise 1%	20	Heterosc.	0.0002 / 0.001	0.0393	69.9	0.040 / 0.043 / 0.056	1/1/3
Transformer	FD004	Noise 1%	20	Fixed var.	0.0005 / 0.003	0.0562	64.3	0.059 / 0.066 / 0.107	4/4/5
Transformer	FD004	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0435	61.4	0.044 / 0.044 / 0.044	2/2/1
Transformer	FD004	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0506	67.7	0.051 / 0.051 / 0.051	3/3/2
Transformer	FD004	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0704	71.8	0.070 / 0.070 / 0.070	5/5/4
Transformer	FD004	Noise 1%	30	Heterosc.	0.0005 / 0.002	0.0707	78.6	0.073 / 0.081 / 0.121	2/2/2
Transformer	FD004	Noise 1%	30	Fixed var.	0.0030 / 0.013	0.0646	78.0	0.080 / 0.125 / 0.368	4/5/5
Transformer	FD004	Noise 1%	30	MC Drop.	0.0030 / 0.013	0.0542	76.4	0.069 / 0.115 / 0.358	1/3/4
Transformer	FD004	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0743	77.9	0.074 / 0.074 / 0.074	3/1/1
Transformer	FD004	Noise 1%	30	CP-norm	0.0008 / 0.004	0.1067	82.5	0.111 / 0.124 / 0.191	5/4/3
Transformer	FD004	Bias 5%	10	Heterosc.	0.0093 / 0.122	0.3200	87.0	0.366 / 0.506 / 1.248	1/2/4
Transformer	FD004	Bias 5%	10	Fixed var.	0.0076 / 0.100	0.3661	87.1	0.404 / 0.518 / 1.126	4/4/2
Transformer	FD004	Bias 5%	10	MC Drop.	0.0103 / 0.136	0.3526	87.1	0.404 / 0.558 / 1.382	5/5/5
Transformer	FD004	Bias 5%	10	Ensemble	0.0034 / 0.044	0.3612	87.5	0.378 / 0.429 / 0.699	2/1/1
Transformer	FD004	Bias 5%	10	CP-norm	0.0084 / 0.111	0.3409	86.8	0.383 / 0.510 / 1.185	3/3/3
Transformer	FD004	Bias 5%	20	Heterosc.	0.0177 / 0.105	0.3186	95.9	0.407 / 0.673 / 2.091	2/4/4
Transformer	FD004	Bias 5%	20	Fixed var.	0.0157 / 0.093	0.3576	95.8	0.436 / 0.672 / 1.927	4/3/3
Transformer	FD004	Bias 5%	20	MC Drop.	0.0187 / 0.111	0.3512	96.0	0.445 / 0.726 / 2.225	5/5/5
Transformer	FD004	Bias 5%	20	Ensemble	0.0059 / 0.035	0.3494	96.8	0.379 / 0.468 / 0.940	1/1/1
Transformer	FD004	Bias 5%	20	CP-norm	0.0147 / 0.087	0.3377	96.2	0.411 / 0.631 / 1.806	3/2/2
Transformer	FD004	Bias 5%	30	Heterosc.	0.0208 / 0.091	0.3328	100.4	0.437 / 0.748 / 2.409	2/3/4
Transformer	FD004	Bias 5%	30	Fixed var.	0.0196 / 0.086	0.3642	100.0	0.462 / 0.756 / 2.322	4/4/3
Transformer	FD004	Bias 5%	30	MC Drop.	0.0221 / 0.097	0.3624	100.3	0.473 / 0.805 / 2.573	5/5/5
Transformer	FD004	Bias 5%	30	Ensemble	0.0084 / 0.037	0.3612	100.4	0.403 / 0.530 / 1.205	1/1/1
Transformer	FD004	Bias 5%	30	CP-norm	0.0181 / 0.079	0.3468	100.1	0.437 / 0.708 / 2.153	3/2/2
Transformer	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.4034	78.3	0.403 / 0.403 / 0.403	1/1/1
Transformer	FD004	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.6768	89.1	0.677 / 0.677 / 0.677	5/5/5
Transformer	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.6439	87.8	0.644 / 0.644 / 0.644	4/4/4
Transformer	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.4557	81.3	0.456 / 0.456 / 0.456	3/3/3
Transformer	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.4498	81.0	0.450 / 0.450 / 0.450	2/2/2
Transformer	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.5249	93.8	0.525 / 0.525 / 0.525	1/1/1
Transformer	FD004	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6591	98.0	0.659 / 0.659 / 0.659	5/5/5
Transformer	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6430	97.4	0.643 / 0.643 / 0.643	4/4/4
Transformer	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.5738	94.8	0.574 / 0.574 / 0.574	3/3/3
Transformer	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.5603	95.2	0.560 / 0.560 / 0.560	2/2/2
Transformer	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6059	100.8	0.606 / 0.606 / 0.606	1/1/1
Transformer	FD004	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6354	101.6	0.635 / 0.635 / 0.635	4/4/4
Transformer	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6329	101.5	0.633 / 0.633 / 0.633	3/3/3
Transformer	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6371	101.9	0.637 / 0.637 / 0.637	5/5/5
Transformer	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6135	100.9	0.614 / 0.614 / 0.614	2/2/2

S2B. RANK REVERSAL ACROSS COST RATIOS

Reversal: the ranking of the five mechanisms differs between at least two of the cost ratios; top change: the top-ranked mechanism differs. Totals: 36 of 96 reversals, 20 top changes.

Backbone	Data	Cond.	L	Reversal	Top change
LSTM	FD001	Clean	10	no	no
LSTM	FD001	Clean	20	yes	no

Backbone	Data	Cond.	L	Reversal	Top change
LSTM	FD001	Clean	30	no	no
LSTM	FD001	Noise 1%	10	no	no
LSTM	FD001	Noise 1%	20	yes	no
LSTM	FD001	Noise 1%	30	yes	no
LSTM	FD001	Bias 5%	10	no	no
LSTM	FD001	Bias 5%	20	yes	no
LSTM	FD001	Bias 5%	30	yes	no
LSTM	FD001	Drift 5%	10	no	no
LSTM	FD001	Drift 5%	20	no	no
LSTM	FD001	Drift 5%	30	no	no
LSTM	FD002	Clean	10	no	no
LSTM	FD002	Clean	20	no	no
LSTM	FD002	Clean	30	no	no
LSTM	FD002	Noise 1%	10	yes	yes
LSTM	FD002	Noise 1%	20	yes	yes
LSTM	FD002	Noise 1%	30	yes	yes
LSTM	FD002	Bias 5%	10	no	no
LSTM	FD002	Bias 5%	20	yes	no
LSTM	FD002	Bias 5%	30	no	no
LSTM	FD002	Drift 5%	10	no	no
LSTM	FD002	Drift 5%	20	no	no
LSTM	FD002	Drift 5%	30	no	no
LSTM	FD003	Clean	10	no	no
LSTM	FD003	Clean	20	no	no
LSTM	FD003	Clean	30	no	no
LSTM	FD003	Noise 1%	10	no	no
LSTM	FD003	Noise 1%	20	no	no
LSTM	FD003	Noise 1%	30	no	no
LSTM	FD003	Bias 5%	10	yes	no
LSTM	FD003	Bias 5%	20	yes	no
LSTM	FD003	Bias 5%	30	yes	yes
LSTM	FD003	Drift 5%	10	no	no
LSTM	FD003	Drift 5%	20	no	no
LSTM	FD003	Drift 5%	30	no	no
LSTM	FD004	Clean	10	yes	yes
LSTM	FD004	Clean	20	yes	yes
LSTM	FD004	Clean	30	yes	yes
LSTM	FD004	Noise 1%	10	yes	yes
LSTM	FD004	Noise 1%	20	yes	yes
LSTM	FD004	Noise 1%	30	yes	yes
LSTM	FD004	Bias 5%	10	yes	yes
LSTM	FD004	Bias 5%	20	yes	no
LSTM	FD004	Bias 5%	30	yes	no
LSTM	FD004	Drift 5%	10	no	no
LSTM	FD004	Drift 5%	20	no	no
LSTM	FD004	Drift 5%	30	no	no
Transformer	FD001	Clean	10	no	no
Transformer	FD001	Clean	20	no	no
Transformer	FD001	Clean	30	no	no
Transformer	FD001	Noise 1%	10	no	no
Transformer	FD001	Noise 1%	20	no	no
Transformer	FD001	Noise 1%	30	no	no
Transformer	FD001	Bias 5%	10	no	no
Transformer	FD001	Bias 5%	20	yes	no
Transformer	FD001	Bias 5%	30	yes	no
Transformer	FD001	Drift 5%	10	no	no
Transformer	FD001	Drift 5%	20	no	no
Transformer	FD001	Drift 5%	30	no	no
Transformer	FD002	Clean	10	yes	yes
Transformer	FD002	Clean	20	yes	yes
Transformer	FD002	Clean	30	no	no
Transformer	FD002	Noise 1%	10	yes	yes
Transformer	FD002	Noise 1%	20	yes	yes
Transformer	FD002	Noise 1%	30	no	no
Transformer	FD002	Bias 5%	10	yes	no
Transformer	FD002	Bias 5%	20	no	no
Transformer	FD002	Bias 5%	30	no	no
Transformer	FD002	Drift 5%	10	no	no
Transformer	FD002	Drift 5%	20	no	no
Transformer	FD002	Drift 5%	30	no	no
Transformer	FD003	Clean	10	no	no
Transformer	FD003	Clean	20	no	no
Transformer	FD003	Clean	30	no	no
Transformer	FD003	Noise 1%	10	no	no
Transformer	FD003	Noise 1%	20	no	no

Backbone	Data	Cond.	L	Reversal	Top change
Transformer	FD003	Noise 1%	30	no	no
Transformer	FD003	Bias 5%	10	yes	no
Transformer	FD003	Bias 5%	20	no	no
Transformer	FD003	Bias 5%	30	no	no
Transformer	FD003	Drift 5%	10	no	no
Transformer	FD003	Drift 5%	20	no	no
Transformer	FD003	Drift 5%	30	no	no
Transformer	FD004	Clean	10	no	no
Transformer	FD004	Clean	20	no	no
Transformer	FD004	Clean	30	yes	yes
Transformer	FD004	Noise 1%	10	yes	yes
Transformer	FD004	Noise 1%	20	yes	yes
Transformer	FD004	Noise 1%	30	yes	yes
Transformer	FD004	Bias 5%	10	yes	yes
Transformer	FD004	Bias 5%	20	yes	no
Transformer	FD004	Bias 5%	30	yes	no
Transformer	FD004	Drift 5%	10	no	no
Transformer	FD004	Drift 5%	20	no	no
Transformer	FD004	Drift 5%	30	no	no

S2C. PER-ENGINE BOOTSTRAP OF TOP-RANKED MECHANISM AGAINST THE OTHERS

For each (condition, L , ratio) the cost difference between the top-ranked mechanism and each of the other four was bootstrapped over engines (2000 resamples). Number of the 144 comparisons per backbone–dataset whose 95% interval excludes zero:

Backbone	Data	Excluding zero
LSTM	FD001	79 / 144
LSTM	FD002	86 / 144
LSTM	FD003	89 / 144
LSTM	FD004	70 / 144
Transformer	FD001	63 / 144
Transformer	FD002	90 / 144
Transformer	FD003	81 / 144
Transformer	FD004	101 / 144

The full 1152 intervals are in the source file.

S3. SINGLE-LEVEL WEIGHTED INTERVAL SCORE

Sources: `results/clean/interval_score_wis_clean.json` (heteroscedastic, ensemble, CP-norm) and `results/clean/fair_calibration_main_table.json` (fixed variance and MC Dropout, calibration-set scale). With one interval level ($\alpha = 0.10$, $K = 1$) the score used is $WIS = \frac{1}{K+1/2} [\frac{1}{2}|y - \hat{\mu}| + \frac{\alpha}{2} IS_\alpha]$, with $\hat{\mu}$ in place of the median. Means over the terminal unit, clean inputs, five seeds (ensemble: single $M = 5$).

Backbone	Data	Heterosc.	Fixed var.	MC Drop.	Ensemble	CP-norm
LSTM	FD001	6.03	6.11	6.04	5.71	6.02
LSTM	FD002	6.16	6.81	6.82	5.83	6.18
LSTM	FD003	5.34	5.74	5.70	4.89	5.33
LSTM	FD004	7.68	7.65	7.64	7.38	7.67
Transformer	FD001	5.68	5.66	5.38	5.29	5.64
Transformer	FD002	6.12	6.42	6.47	5.80	6.14
Transformer	FD003	5.47	5.69	5.33	4.89	5.52
Transformer	FD004	7.19	7.31	7.36	6.66	7.20

S4. LATENCY DISTRIBUTIONS

Source: `results/latency/lstm_latency.json`. Per-sample latency in microseconds, median (IQR) over eleven repetitions, one NVIDIA RTX A5000 Laptop GPU; no repetition excluded. *CUDA-event timing*

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop. ($T=50$)	Ensemble ($M=5$)
FD001	LSTM	1	553.89 (71.65)	402.37 (24.66)	527.46 (29.74)	13801.47 (1039.97)	1761.28 (91.54)

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop. ($T=50$)	Ensemble ($M=5$)
FD001	LSTM	32	10.98 (1.04)	9.69 (0.32)	10.56 (0.97)	436.10 (16.21)	44.87 (1.34)
FD001	LSTM	512	1.80 (0.12)	1.68 (0.08)	1.72 (0.08)	89.64 (1.10)	8.02 (0.22)
FD001	Transformer	1	802.82 (67.09)	717.82 (37.25)	798.66 (43.89)	48316.42 (1233.41)	3516.42 (57.34)
FD001	Transformer	32	21.98 (1.88)	20.61 (1.22)	23.20 (1.21)	1940.29 (64.85)	113.95 (3.30)
FD001	Transformer	512	5.22 (0.09)	5.69 (0.11)	5.26 (0.05)	420.23 (0.40)	35.33 (0.18)
FD002	LSTM	1	364.54 (46.51)	324.54 (32.78)	366.59 (28.05)	15504.38 (355.22)	1605.63 (78.40)
FD002	LSTM	32	12.77 (3.09)	10.50 (1.09)	11.93 (0.61)	498.24 (11.05)	56.42 (1.86)
FD002	LSTM	512	2.80 (0.06)	2.73 (0.04)	2.78 (0.07)	140.70 (0.81)	13.23 (0.07)
FD002	Transformer	1	827.39 (76.24)	761.86 (87.02)	812.03 (98.37)	58784.77 (1043.97)	3904.51 (136.19)
FD002	Transformer	32	25.63 (2.13)	24.96 (2.18)	25.38 (1.40)	2109.92 (82.58)	127.04 (3.76)
FD002	Transformer	512	27.94 (1.07)	25.65 (1.24)	28.07 (0.54)	442.88 (40.97)	36.61 (0.04)
FD003	LSTM	1	351.23 (13.31)	318.46 (20.48)	335.87 (29.70)	14606.34 (1037.82)	1685.50 (80.42)
FD003	LSTM	32	12.13 (1.17)	9.50 (0.44)	11.55 (0.58)	487.29 (10.01)	55.74 (3.15)
FD003	LSTM	512	2.81 (0.04)	2.80 (0.05)	2.90 (0.06)	140.14 (0.48)	13.07 (0.20)
FD003	Transformer	1	755.71 (126.98)	681.09 (123.94)	739.33 (58.82)	57734.14 (2224.64)	3513.34 (39.94)
FD003	Transformer	32	25.32 (1.04)	22.21 (1.38)	25.38 (1.14)	1828.22 (58.99)	105.38 (3.37)
FD003	Transformer	512	7.06 (0.05)	7.04 (0.05)	7.16 (0.11)	418.41 (0.56)	35.34 (0.34)
FD004	LSTM	1	355.33 (24.03)	282.62 (9.30)	347.14 (13.79)	15542.27 (262.64)	1697.79 (80.74)
FD004	LSTM	32	11.07 (1.14)	8.61 (0.64)	10.78 (1.15)	493.25 (17.49)	51.46 (2.17)
FD004	LSTM	512	2.82 (0.08)	2.79 (0.08)	2.80 (0.08)	139.76 (0.96)	12.97 (0.14)
FD004	Transformer	1	887.81 (86.02)	786.43 (84.18)	798.69 (46.51)	57554.94 (26801.15)	4062.21 (237.57)
FD004	Transformer	32	25.25 (2.43)	20.49 (0.99)	22.18 (0.42)	2005.98 (124.10)	117.09 (3.34)
FD004	Transformer	512	6.15 (0.11)	6.34 (0.29)	6.27 (0.17)	419.99 (0.38)	35.56 (0.10)

Wall-clock timing

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop. ($T=50$)	Ensemble ($M=5$)
FD001	LSTM	1	500.51 (24.31)	359.83 (38.64)	542.25 (54.12)	15656.49 (786.87)	1717.93 (124.91)
FD001	LSTM	32	10.41 (0.58)	8.72 (0.68)	10.06 (0.33)	435.96 (11.28)	43.46 (1.91)
FD001	LSTM	512	1.74 (0.05)	1.70 (0.03)	1.74 (0.02)	87.15 (1.46)	7.94 (0.16)
FD001	Transformer	1	749.24 (47.45)	699.36 (13.40)	747.39 (22.79)	52755.86 (3971.31)	3498.63 (82.77)
FD001	Transformer	32	22.17 (2.00)	21.08 (1.94)	21.70 (1.11)	2063.85 (139.81)	110.71 (4.05)
FD001	Transformer	512	5.22 (0.13)	5.75 (0.02)	5.36 (0.06)	420.83 (0.70)	35.46 (0.10)
FD002	LSTM	1	368.77 (15.24)	303.89 (20.86)	360.94 (35.11)	15178.14 (1757.13)	1603.89 (48.64)
FD002	LSTM	32	12.20 (0.66)	10.30 (1.23)	12.05 (0.53)	503.28 (7.08)	57.77 (1.51)
FD002	LSTM	512	2.90 (0.01)	2.86 (0.02)	2.90 (0.02)	143.09 (1.22)	13.40 (0.05)
FD002	Transformer	1	857.85 (77.62)	739.60 (37.45)	839.28 (75.11)	55607.90 (3107.32)	3703.63 (223.77)
FD002	Transformer	32	25.37 (1.80)	24.21 (3.85)	24.88 (2.09)	2053.94 (259.80)	126.65 (2.93)
FD002	Transformer	512	29.50 (0.76)	29.11 (2.66)	29.59 (1.25)	433.07 (2.95)	36.69 (0.32)
FD003	LSTM	1	359.95 (13.19)	333.89 (14.81)	361.56 (22.88)	15153.77 (963.71)	1732.46 (55.40)
FD003	LSTM	32	11.26 (0.70)	9.20 (0.57)	12.08 (7.44)	487.54 (20.19)	59.35 (3.80)
FD003	LSTM	512	2.87 (0.05)	2.91 (0.09)	3.01 (0.14)	140.99 (6.00)	12.57 (0.09)
FD003	Transformer	1	702.29 (54.09)	619.36 (25.58)	691.11 (46.38)	57821.99 (1640.56)	3497.32 (82.45)
FD003	Transformer	32	23.94 (0.67)	21.36 (1.88)	24.68 (2.19)	1833.62 (120.30)	108.21 (7.14)
FD003	Transformer	512	7.22 (0.05)	7.18 (0.05)	7.20 (0.06)	417.44 (0.40)	35.69 (0.60)
FD004	LSTM	1	311.82 (26.08)	268.99 (30.87)	324.69 (34.08)	14728.66 (802.19)	1643.36 (38.18)
FD004	LSTM	32	9.78 (0.90)	8.66 (0.44)	9.74 (0.33)	512.11 (16.48)	53.01 (3.46)
FD004	LSTM	512	2.89 (0.09)	2.88 (0.10)	2.82 (0.13)	137.89 (0.87)	12.99 (0.21)
FD004	Transformer	1	783.19 (61.40)	705.95 (50.81)	791.37 (48.03)	54753.97 (2731.39)	4268.16 (406.60)
FD004	Transformer	32	22.22 (0.97)	21.35 (2.48)	21.25 (1.22)	2047.53 (114.06)	116.07 (1.92)
FD004	Transformer	512	6.14 (0.04)	7.02 (0.14)	6.33 (0.16)	412.15 (0.27)	35.12 (0.15)